GLUING SOLUTIONS IN THE FINITE CASE: EXISTENCE AND UNIQUENESS UP TO UNIQUE ISOMORPHISM

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Allowed standard prerequisites (editorial).

- delivered companion: gluing lemma. delivered companion lemma-glue (S11-csh-glue). Produces the glued object K with compatible isomorphisms in the n=2 case.
- Same-chapter uniqueness lemma and remark. Cohomology, Lemma cohomology-lemma-uniqueness; Remark cohomology-remark-uniqueness. Uniqueness up to unique isomorphism; compatibility property pinning the identifications.
- Standing situation (statement level). Cohomology, Situation cohomology-situation-locally-given. Locally given system (K_U, rho) with gluing isomorphisms on a basis.

Source reference. Original reference 0D67: sources/stacks-chow/tags.txt, source lines 16512–16512. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0D68: sources/stacks-chow/tags.txt, source lines 16513–16513. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original remark 0D69: sources/stacks-chow/tags.txt, source lines 16514–16514. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 08DG: sources/stacks-chow/tags.txt, source lines 10636–10636. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *In Situation 0D67 assume*

(1) $X = U_1 \cup \ldots \cup U_n$ *with* $U_i \in \mathcal{B}$,
(2) *for* $U, V \in \mathcal{B}$ *we have* $U \cap V = \bigcup_{W \in \mathcal{B}, W \subset U \cap V} W$,
(3) *for any* $U \in \mathcal{B}$ *we have* $\mathrm{Ext}^i(K_U, K_U) = 0$ *for* $i < 0$.

Then a solution exists and is unique up to unique isomorphism.

Proof. Uniqueness was seen in Lemma 0D68. We may prove the lemma by induction on n. The case $n = 1$ is immediate.

The case $n = 2$. Consider the isomorphism $\rho_{U_1,U_2} : K_{U_1}|_{U_1 \cap U_2} \to K_{U_2}|_{U_1 \cap U_2}$ constructed in Remark 0D69. By Lemma 08DG we obtain an object K in $D(\mathcal{O}_X)$ and isomorphisms $\rho_{U_1} : K|_{U_1} \to K_{U_1}$ and $\rho_{U_2} : K|_{U_2} \to K_{U_2}$ compatible with ρ_{U_1,U_2}. Take $U \in \mathcal{B}$. We will construct an isomorphism $\rho_U : K|_U \to K_U$ and we

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will leave it to the reader to verify that (K, ρ_U) is a solution. Consider the set $\mathcal{B}'$ of elements of $\mathcal{B}$ contained in either $U \cap U_1$ or contained in $U \cap U_2$. Then $(K_U, \rho^U_{U'})$ is a solution for the system $(\{K_{U'}\}_{U' \in \mathcal{B}'}, \{\rho^{U'}_{V'}\}_{V' \subset U' \text{ with } U', V' \in \mathcal{B}'})$ on the ringed space U. We claim that $(K|_U, \tau_{U'})$ is another solution where $\tau_{U'}$ for $U' \in \mathcal{B}'$ is chosen as follows: if $U' \subset U_1$ then we take the composition

$$K|_{U'} \xrightarrow{\rho_{U_1}|_{U'}} K_{U_1}|_{U'} \xrightarrow{\rho^{U_1}_{U'}} K_{U'}$$

and if $U' \subset U_2$ then we take the composition

$$K|_{U'} \xrightarrow{\rho_{U_2}|_{U'}} K_{U_2}|_{U'} \xrightarrow{\rho^{U_2}_{U'}} K_{U'}.$$

To verify this is a solution use the property of the map ρ_{U_1, U_2} described in Remark 0D69 and the compatibility of ρ_{U_1} and ρ_{U_2} with ρ_{U_1, U_2}. Having said this we apply Lemma 0D68 to see that we obtain a unique isomorphism $K|_{U'} \to K_{U'}$ compatible with the maps $\tau_{U'}$ and $\rho^U_{U'}$ for $U' \in \mathcal{B}'$.

The case $n > 2$. Consider the open subspace $X' = U_1 \cup \ldots \cup U_{n-1}$ and let $\mathcal{B}'$ be the set of elements of $\mathcal{B}$ contained in X'. Then we find a system $(\{K_U\}_{U \in \mathcal{B}'}, \{\rho^U_V\}_{U, V \in \mathcal{B}'})$ on the ringed space X' to which we may apply our induction hypothesis. We find a solution $(K_{X'}, \rho^{X'}_U)$. Then we can consider the collection $\mathcal{B}^* = \mathcal{B} \cup \{X'\}$ of opens of X and we see that we obtain a system $(\{K_U\}_{U \in \mathcal{B}^*}, \{\rho^U_V\}_{V \subset U \text{ with } U, V \in \mathcal{B}^*})$. Note that this new system also satisfies condition (3) by Lemma 0D68 applied to the solution $K_{X'}$. For this system we have $X = X' \cup U_n$. This reduces us to the case $n = 2$ we worked out above. $\square$